Loan Approval Prediction

Data Science Internship Project — Model Evaluation Report

1. Project Objective

The objective is to build and evaluate a supervised machine-learning model that predicts whether a loan application is likely to be approved. The model is intended to support review and decision-making, not replace human judgement or lending policies.

2. Dataset and Preparation

The dataset contained 614 records and 13 columns, including applicant characteristics, income, loan details, credit history, property area, and loan status. Target labels were standardized: approved (Y) = 1 and not approved (N) = 0. Duplicate rows were removed. Numeric features used median imputation and standard scaling; categorical features used most-frequent imputation and one-hot encoding. Loan ID was excluded as an identifier. A stratified train/validation/test split was used.

3. Models Compared

Logistic Regression and Random Forest were evaluated. Class weights were balanced to help address the unequal class counts. The decision threshold was selected using validation data by comparing F1-scores over a range of thresholds.

4. Final Test Results

Metric	Result
Selected model	Logistic Regression
Selected threshold	0.20
Accuracy	0.854
Precision (approved class)	0.832
Recall (approved class)	0.988
F1-score (approved class)	0.903
ROC-AUC	0.856

On the test set, class 0 (not approved) had precision 0.95, recall 0.55, and F1-score 0.70 (38 cases). Class 1 (approved) had precision 0.83, recall 0.99, and F1-score 0.90 (85 cases). Overall accuracy was approximately 0.85.

5. Interpretation and Trade-offs

At threshold 0.20, the model identifies nearly all applications that were approved in the test set, contributing to high recall for class 1. However, recall for class 0 is 0.55, so a substantial share of not-approved cases were predicted as approved. In lending, this could increase credit risk if predictions were used directly to approve loans. Maximizing validation F1-score does not necessarily minimize financial loss or ensure fairness.

6. Recommendations

• Use the model as decision support and retain human review and lending rules.
• Choose a deployment threshold based on the costs of false approvals and false rejections, not F1-score alone.
• Review false positives and false negatives and assess performance across relevant applicant groups for potential bias.
• Validate on additional, recent data and monitor performance drift before deployment.
• Document decisions and provide appropriate review processes for affected applicants.

7. Conclusion

Logistic Regression was selected by the validation threshold search and achieved test ROC-AUC 0.856 and approved-class F1-score 0.903 at threshold 0.20. The results show useful predictive signal, but low recall for the not-approved class is an important limitation. Further threshold analysis, risk evaluation, fairness checks, and human oversight are needed before real-world use.

Note: Results are based on metrics supplied from the project notebook. This model is not a substitute for a full credit-risk assessment.